COMPUTER VISION · MODE: STARTUP

# CarDoctor

*See the damage. Know what to do.*
*Find who can fix it.*


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Data · measurements

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Knowledge base · training


Team MDB Vision · try it: abduloxunov-cardoctor.hf.space



# Buyers and fleets decide on guesswork

**9,364**

road crashes in Uzbekistan in 2024. Each one needs a damage check

Kun.uz, Feb 2025

**~240k**

used cars sold in H1 2026, and parliament is now fining odometer fraud

Kun.uz, 2026

**2 of 3**

drivers don't trust repair shops (unneeded services, overcharging)

AAA survey



### Used-car buyers and sellers

Can't see hidden damage before the visit; the seller won't say.



### Truck and taxi fleets

A breakdown on the road: stop or keep driving? Who should we call?

# Live demo: pick a car, or use your own photo

1

## Sell / Buy

Sample car → verdict, damage boxes on every photo, 3D car with red areas, report

2

## Fleet & trucks

"Cobalt, engine overheating, steam" → 🔴 Stop and get help, with the manual source

3

## Route

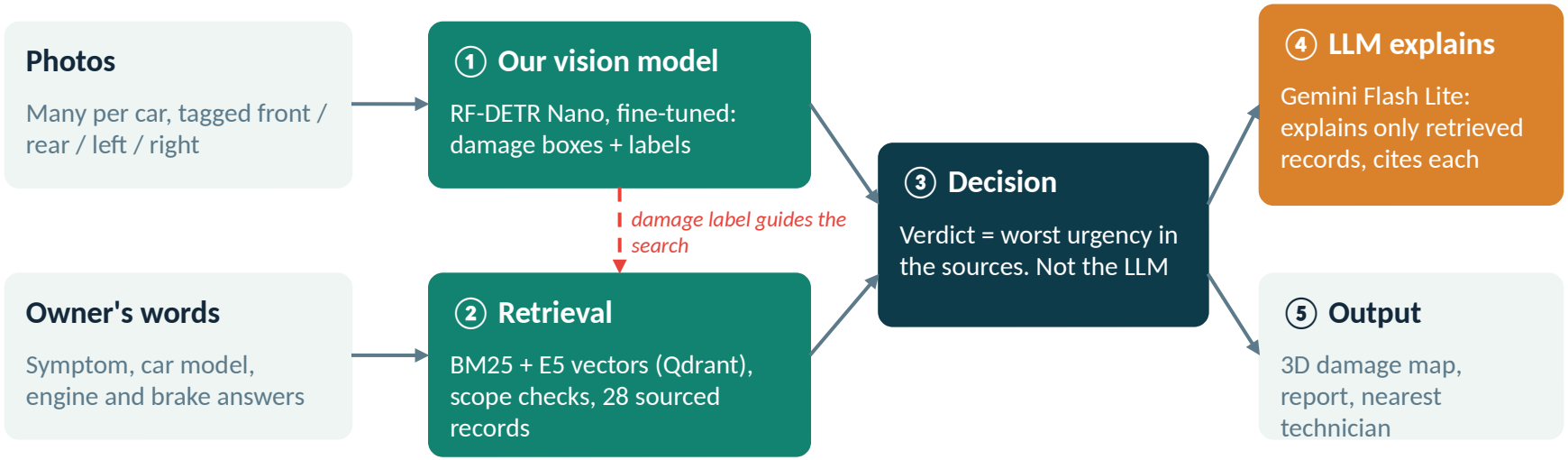
Use my location → nearest technician + route; real workshops on Google Maps



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Open it on your phone. Sample 1, 2 or 3, or upload your own photos.

# How the model and the LLM work together




## Vision

sees what is visible

## Knowledge base

knows what it means and how urgent

## LLM

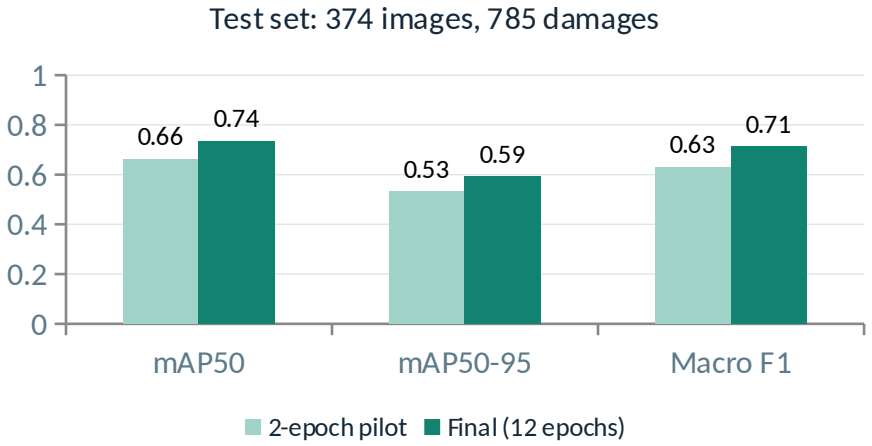
puts it in plain words: it never decides, never sees photos

# We trained it and measured it ourselves

**0.735** mAP50 on our held-out test set

**0.714** macro F1 (conf 0.35, IoU 0.5)

**0.36 s** per photo on a laptop CPU, no GPU




Data: CarDD, 2,816 train / 810 validation images, 6 classes; trained on NVIDIA L4 (Hugging Face Jobs).
Known failure cases: cracks and small defects; not yet tested at night, in rain or on Uzbek cars; a photo cannot show hidden damage.

# Why another team can't rebuild it in a weekend

1

## Own detector

Fine-tuned RF-DETR with our training and evaluation pipeline and per-class results

2

## Local knowledge

Sourced records for Cobalt, Lacetti, Nexia and Damas, with manual page references

3

## Safety logic

Asks "engine running? parking brake released?" before model-specific advice

4

## Next data moat

Our photos of Uzbek cars + confirmed repair outcomes from partner workshops

# One report, three paying sides

### Marketplaces

per condition report or per listing

### Fleets

per vehicle per month

### Workshops

per referred job

## Cost per inspection

CPU time (0.36 s × photos, measured)
+ one optional LLM call (text only)
No GPU needed to run it.

## 3-month pilot

One marketplace or one fleet in Tashkent
Team of 3 + one CPU server
Collect local photos → per-condition accuracy
Measure: minutes saved per inspection

# What comes next

*Local cars, exact parts, real partners*

1. Test set of Uzbek cars at night and in rain
2. Exact-part marking on the 3D car (part segmentation)
3. Verified partner workshops with prices
4. Uzbek and Russian interface

Thank you · abduloxunov-cardoctor.hf.space · github.com/Abduloxunov/mdb-vision-finals

